



OBJECT ORIENTED SOFTWARE ENGINEERING

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Regression Testing

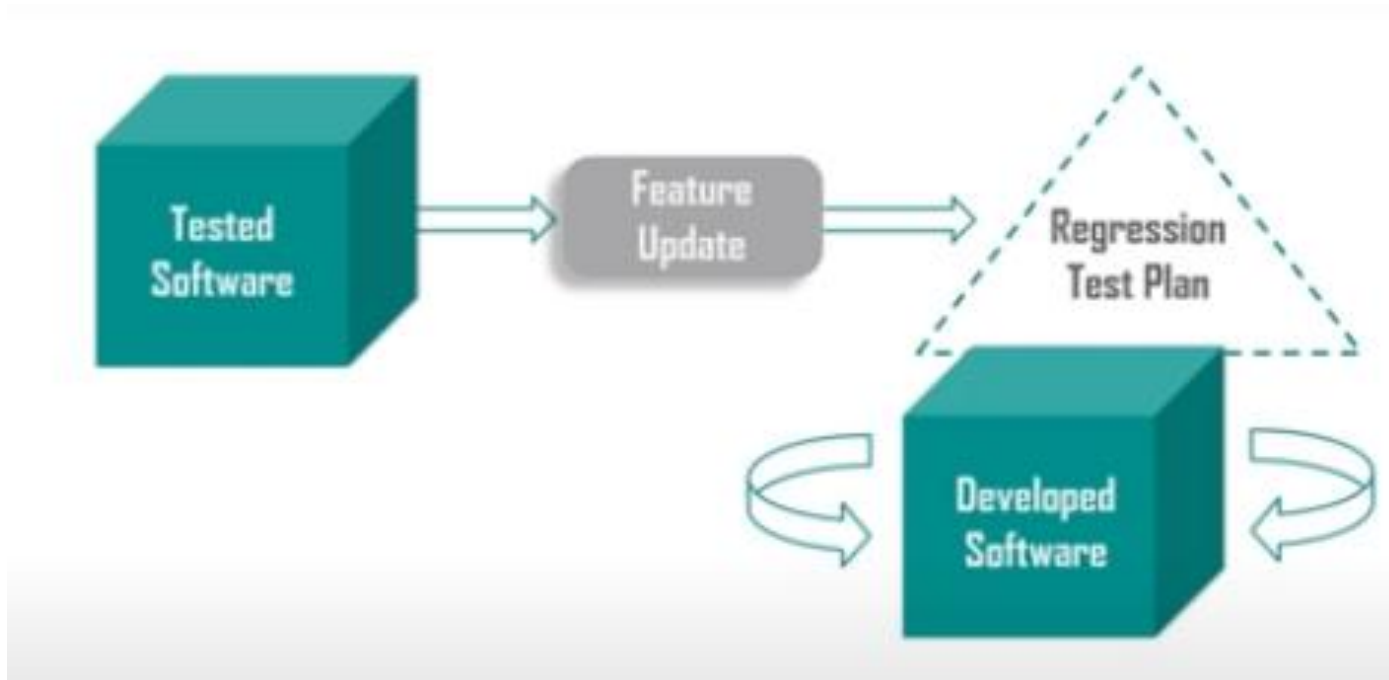
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- Testing of a previously tested programs following modification, to **ensure that defects have not been introduced** or **uncovered in unchanged areas of software**, as a result of the changes made.
- That is, Regression testing **verifies** that **recent code changes haven't altered or destroyed the already existing functionality of a system.**

OBJECT ORIENTED SOFTWARE ENGINEERING

Regression Testing



- Each **new addition** or **change** to baselined software may **cause problems** with functions that **previously worked flawlessly**
- Regression testing re-executes a **small subset of tests** that have **already been conducted**
 - Ensures that changes have **not propagated unintended side effects**
 - Helps to ensure that **changes do not introduce unintended behavior** or **additional errors**
 - May be done manually or through the use of automated capture/playback tools

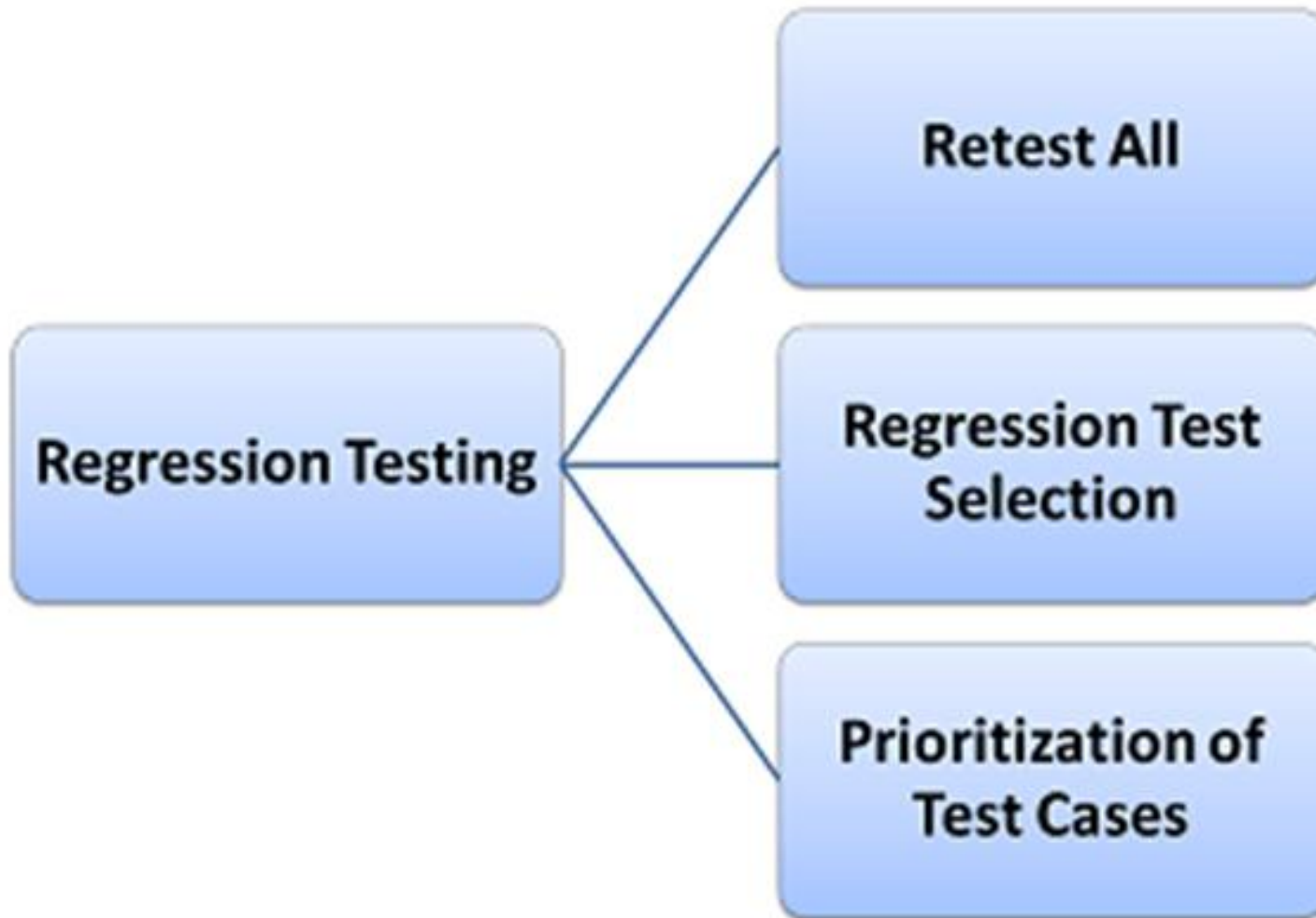
Why Regression Testing ?

- The **Need of Regression Testing** mainly arises whenever there is **requirement to change the code** and we need to **test whether the modified code affects the other part of software application or not**.
- Moreover, regression testing is needed, **when a new feature is added to the software application** and **for defect fixing as well as performance issue fixing**.

How Regression Testing performed?

- In order to do Regression Testing process, we need to **first debug the code** to identify the bugs.
- Once the bugs are identified, required changes are made to fix it, then the regression testing is done by **selecting relevant test cases from the test suite** that covers **both modified and affected parts of the code..**

- Software maintenance is an activity which includes enhancements, error corrections, optimization and deletion of existing features. These modifications may cause the system to work incorrectly.
- Therefore, Regression Testing becomes necessary.
- Regression Testing can be carried out using the following **techniques**:
 - **Retest all**
 - **Regression test selection**
 - **Prioritization of test cases**



1. Retest All:

- This is one of the methods for Regression Testing in which all the tests in the existing test bucket or suite should be **re-executed**. This is very expensive as it requires huge time and resources.

2. Regression Test Selection:

- Regression Test Selection is a technique in which **some selected test cases from test suite** are executed to test whether the modified code affects the software application or not.
- Test cases are categorized into two parts,
 - **reusable test cases** which can be used in further regression cycles and
 - **obsolete test cases** which can not be used in succeeding cycles.

3. Prioritization of Test Cases:

- Prioritize the test cases depending on business impact, critical & frequently used functionalities.
- Selection of test cases based on priority will greatly reduce the regression test suite.

Following are the most important tools used for both functional and regression testing in OBJECT ORIENTED SOFTWARE ENGINEERING:

- **Selenium**: This is an open source tool used for automating web applications. Selenium can be used for browser-based regression testing.
- **Quick Test Professional (QTP)**: HP Quick Test Professional is automated software designed to automate functional and regression test cases. It uses VBScript language for automation. It is a Data-driven, Keyword based tool.
- **Rational Functional Tester (RFT)**: IBM's rational functional tester is a Java tool used to automate the test cases of software applications. This is primarily used for automating regression test cases and it also integrates with Rational Test Manager.



THANK YOU

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